

- 5 A student investigated the effects of air movement on the rate of transpiration by using a potometer.
- (a) Define the term transpiration.

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The potometer used by the student is shown in Fig. 5.1.

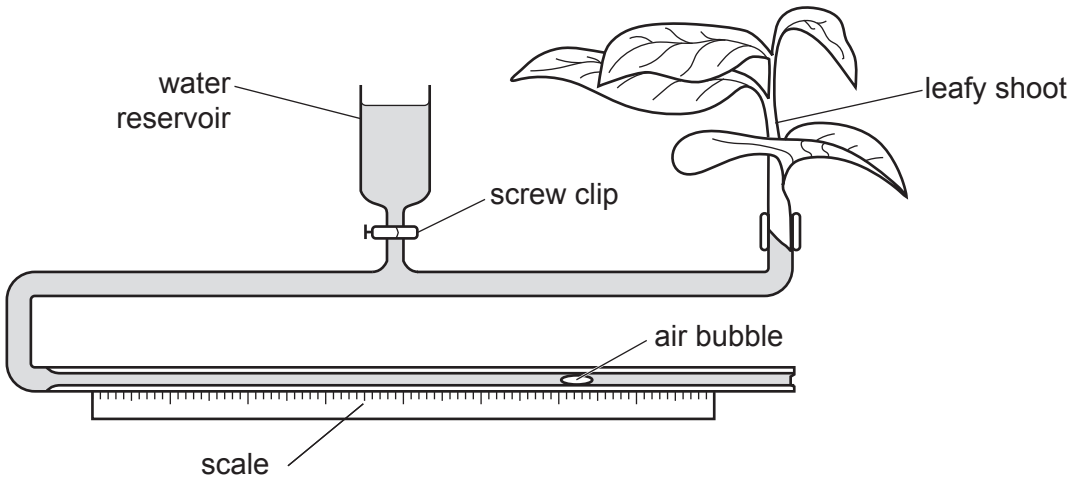


Fig. 5.1

The student recorded the distance the air bubble in the apparatus had moved after ten minutes with no air movement.

The rate of water uptake was used as a measure of the rate of transpiration.

The apparatus was then reset using the reservoir, and the experiment was repeated a further two times. All other variables were standardised during the three experiments.

Table 5.1 shows the results recorded by the student.

Table 5.1

experiment	distance moved by the bubble in 10 minutes/mm
1	12.5
2	12.0
3	11.5

(b) Use Table 5.1 to calculate the mean rate of movement of the bubble.

..... mm min⁻¹ [1]

(c) The student carried out another experiment using the same plant. In this experiment a fan was used to blow air across the leaves of the plant. All other variables were standardised.

The results showed that the bubble moved further in 10 minutes.

The student concluded that air movement increases the rate of transpiration.

Explain why air movement increases the rate of transpiration.

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(d) Explain how water moves up through a xylem vessel in the stem of the plant in the potometer, shown in Fig. 5.1.

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[Total: 9]